Corporate Finance – Business Valuation (Course 20127)

Valuation Multiples

Milan, November 2nd-3rd 2026

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Plan for the day

Last time

- Understanding FCFE Discount Models
- Learning the Differences between DDM and FCFE

Today

- Main questions for today:
  - Understanding the principles of Relative Valuation
  - Learning the basic steps to Using Common Multiples

For the Next Time

- Review the material covered in class

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Agenda

1. Use of Relative Valuation

a. Overview

b. Principles

c. Why Relative Valuation

2. Using Multiples

a. Metrics

b. Steps

c. Deal Multiples

3. Common Multiples

a. P/E

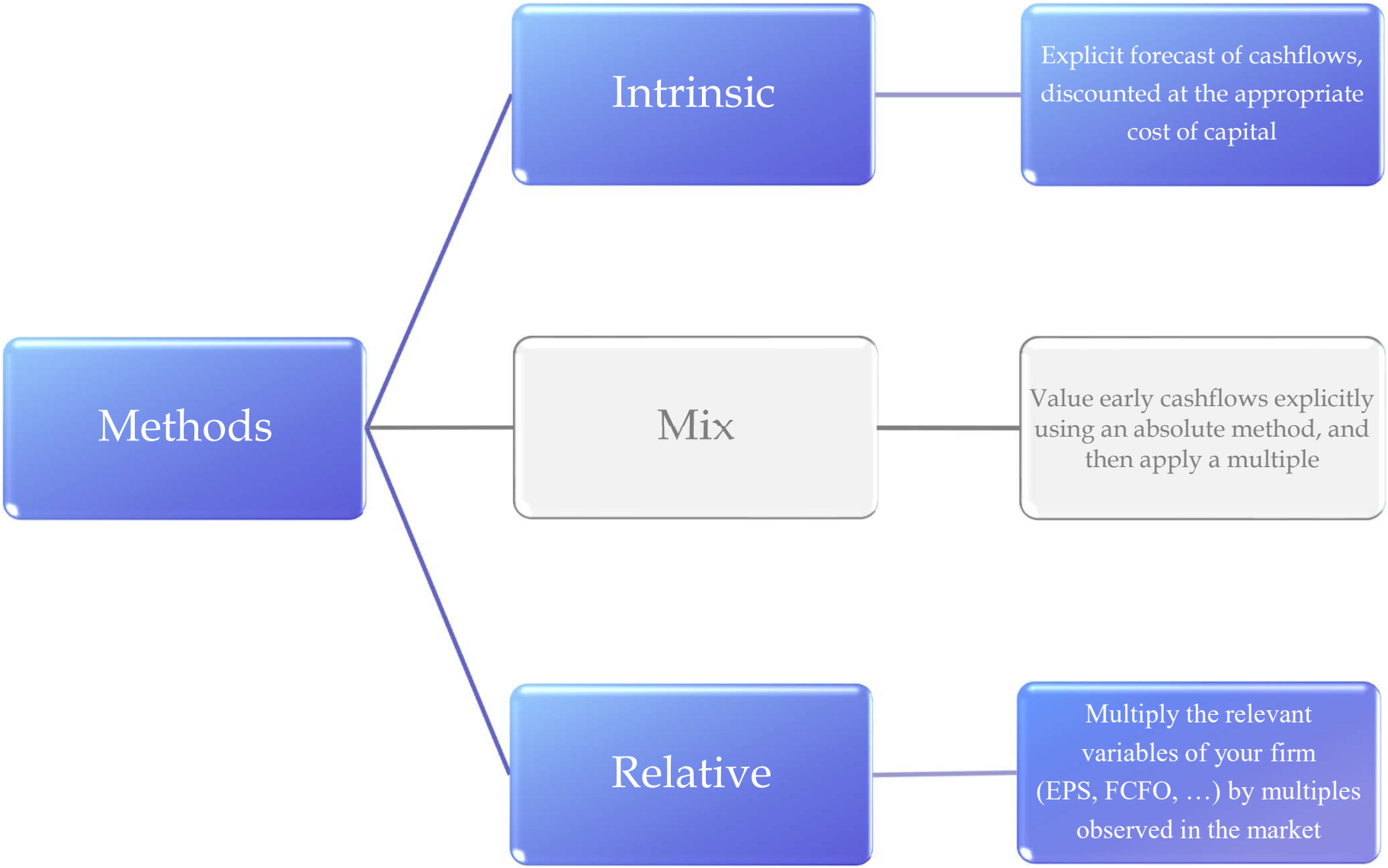
b. P/BV

c. EV/EBITDA

d. EV/Sales

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Relative Valuation: Overview (Reminder)



Relative Valuation: Principles

- **In a relative valuation, the value of an asset is compared to the values assessed by the market for comparable assets**

- **There are two basic components to relative valuations:**

  1. **Standardized prices**

     Prices must be standardized converting them into multiples of Earnings (P/E), BV (P/BV), Sales (EV/Sales) and so forth (so that they're independent of size)

  2. **Similar or comparable firms**

     Need for similar firms (peers). **No two firms are identical** and firms in the same business can still differ on risk, growth and cashflows

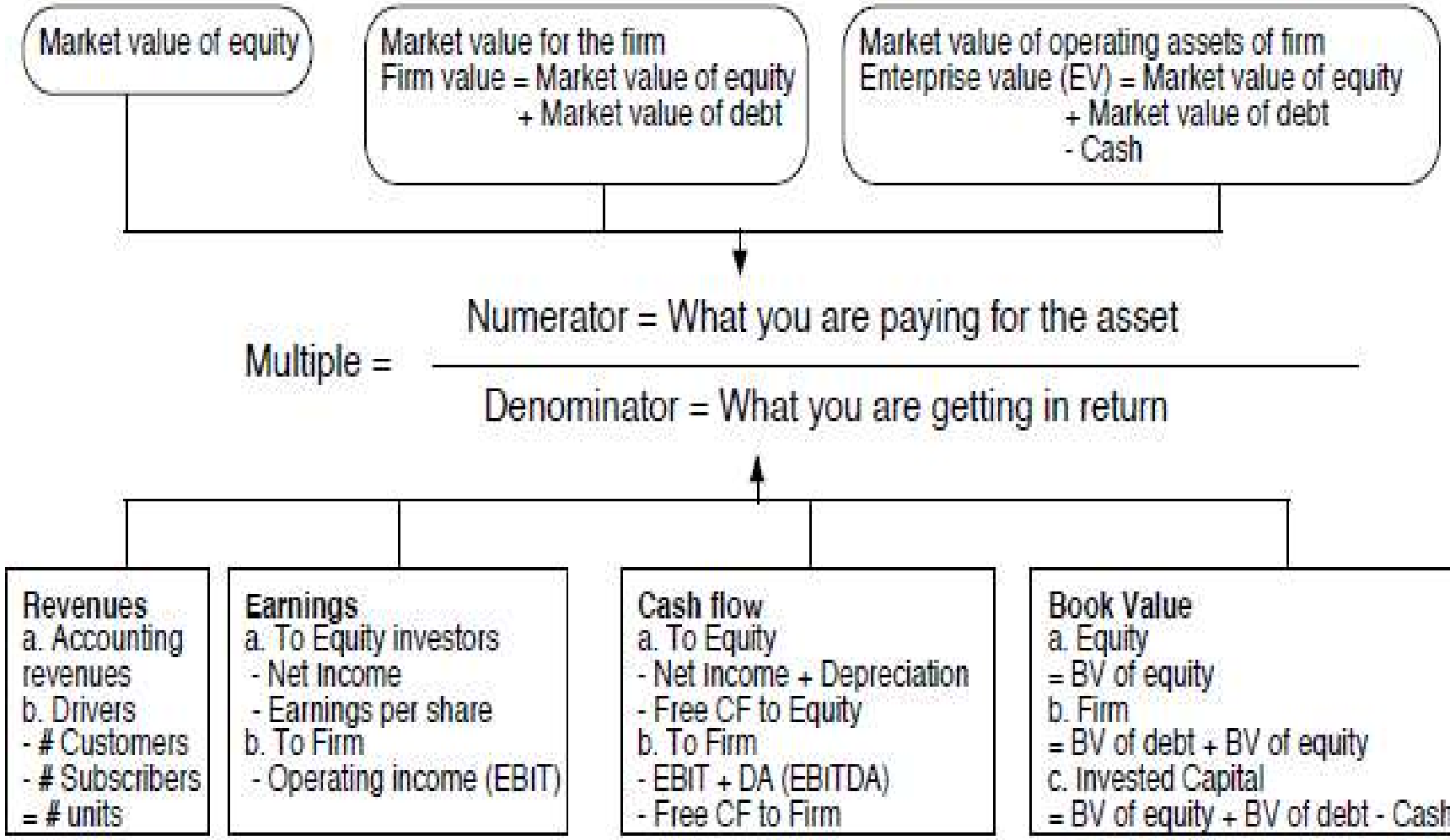
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Relative Valuation: Principles

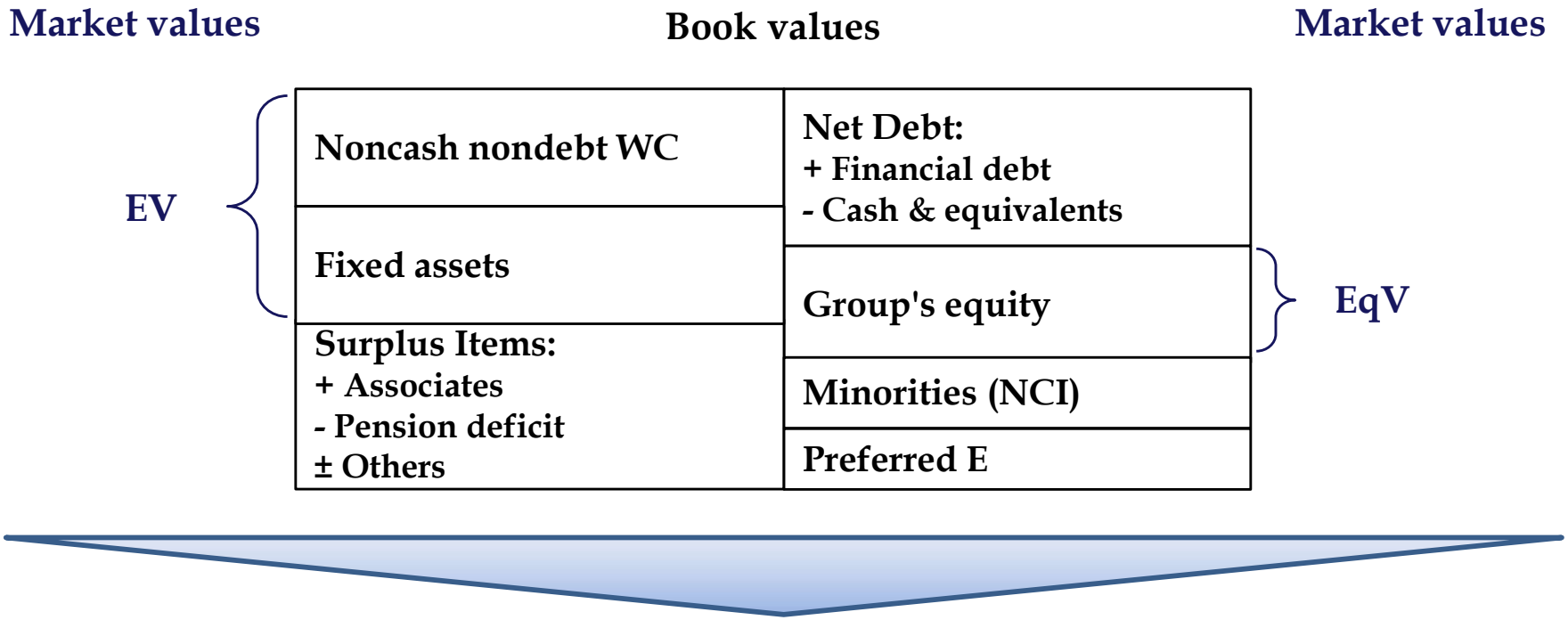
- **Multiples can be sourced from two markets:**
  - ✓ **Stock Market (Market or Trading Multiples)**. They refer to stand-alone values (no acquisition premiums or minority discounts embedded)
  - ✓ **M&A Market (Deal or Transaction Multiples)**. They refer to prices paid in M&A deals (acquisition premiums or minority discounts embedded)
- **Ignoring speculative phenomena (bubbles, …), the difference between market multiples and deal multiples are explained by acquisition premiums or minority discounts**

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Relative Valuation: Principles



Relative Valuation: Principles



- **Moving from EV to EqV (Bridge-to-Equity):**

EV	-	Net debt	+	Assoc.	-	Pension deficit	-	NCI	-	Preferred shares	±	Surplus Items	=	EqV

Relative Valuation: Why?

- **Several reasons why relative valuations are widely used:**
  1. Few explicit assumptions. Far fewer explicit assumptions and far quicker than DCF valuations
  2. Market-oriented. More likely to reflect the current market mood, since they are an attempt to measure relative (not intrinsic) value
  3. Simplicity. Simpler to understand and easier to present than a DCF (easier to frame an asset as cheap or expensive using a multiple)

- **Most valuations on Wall Street are relative valuations:**
  - ✓ Almost all equity reports are based on multiples and comparables
  - ✓ Almost all acquisition valuations are based on multiples
  - ✓ Rules of thumb based on multiples are not only common but often the basis for final valuation judgments

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Relative Valuation: Why?

- **The strengths of relative valuations are also their major weaknesses:**
  1. Few explicit assumptions. The ease with which relative valuations can be done can also result in inconsistent estimates where key variables such as risk, growth or cashflows are ignored

  2. Market oriented. Using multiples can result in values that are too high/low when the market is over/undervaluing comparables

  3. Simplicity (*bias*). The lack of transparency regarding the underlying assumptions makes multiples vulnerable to manipulation

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Using Multiples: Metrics

- **Suppose it's 9/30/2026. The denominator can either be:**
  - ✓ **Actual**: LFY (12/31/2025)
  - ✓ **Forward**: next FY (2026) or following (2027, …)
  - ✓ **Trailing**: last four quarters (LTM): Q4 2015 + Q1/Q2/Q3 2026

- **Time frame choice:**
  - ✓ **LFY/LTM** if no estimates exist (for either comparables or target)
  - ✓ **F+1, +2, …** good if LFY/LTM < 0 and/or for growing/innovative businesses

- **Usually, firms' business plans have increasing forecasts:**
  - ✓ EBITDA 27E > EBITDA 26A → FWD Multiple < ACT Multiple

- **The numerator never changes! Future EV and EqV don't exist**

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Using Multiples: Metrics

- **Using a FWD (T + 1) EBITDA multiple, the target company's EqV as of the end of 2025 is:**

	2025	2026	2027
	Peer Company		
Market Cap	1,000		
EBITDA	100	120	150
Net Debt	200	205	215
	Target Company		
Equity Value	?		
EBITDA	20	25	30
Net Debt	10	15	20

- ✓ 241
- ✓ 235
- ✓ 240
- ✓ 236

Using Multiples: Metrics

- **Choosing a FWD multiple requires consistency between peers and the target firm:**

	Mkt Cap	NI 25A	NI 26E	P/E Actual	P/E FWD
Peers	10,000	550	605	10,000/550 = 18.2x	10,000/605= 16.5x
Target		190	240		

Equity value calculation as of the end of 2025

With P/E Actual **190 * 18.2x = 3,455**

With P/E FWD **240 * 16.5x = 3,967**

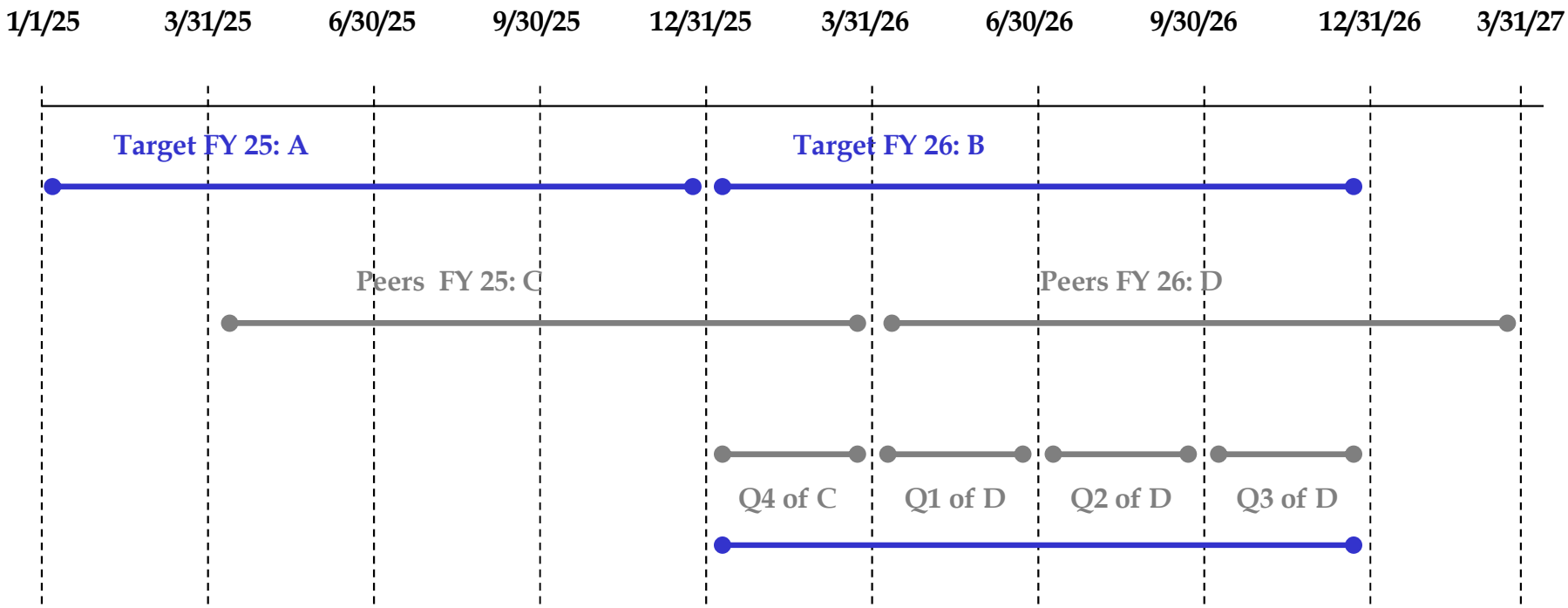
~~**Incorrect**~~ ~~**190 * 16.5x = 3,140**~~

~~**Incorrect**~~ ~~**240 * 18.2x = 4,364**~~

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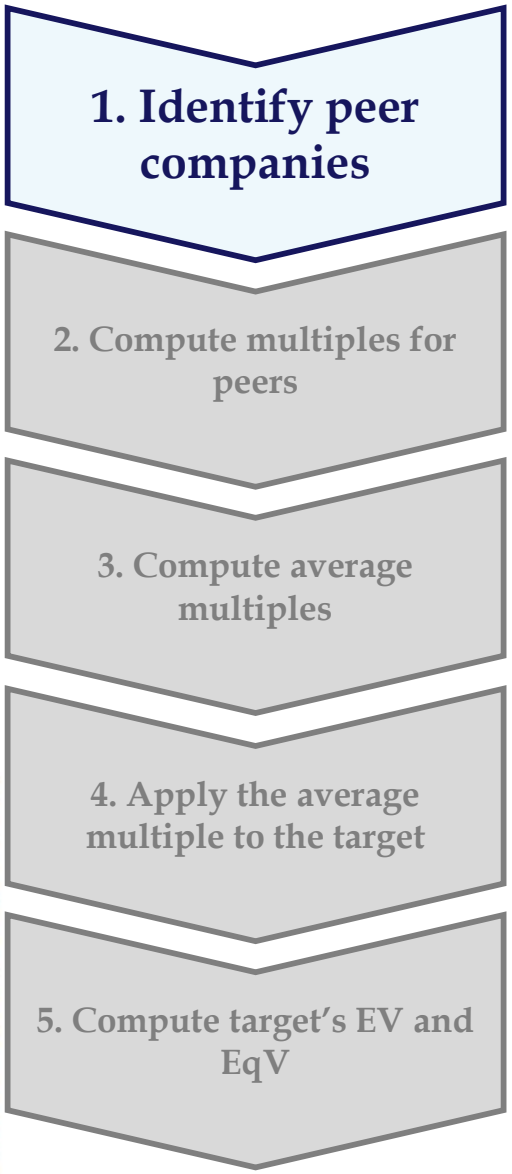
Using Multiples: Metrics

- **Assume target's FY ends on 12/31, and peers' FY ends on 3/31**
- **Peers' figures must be calendarized as of 12/31**



- **Calendarized EBITDA for peers is equal to Q4 (25) + Q1-Q2-Q3 (26)**

Using Multiples: Steps



Comparability Criteria:

Business model

- Activities, products, customers, …
- Operating risk: capital intensity (manufacturing vs outsourcing)
- Growth prospects and stage in the business life cycle

Size

- Absolute values of Sales, EBITDA, market share, …

Margins

- Similar EBITDA % and NI %

Debt

- Similar financial risk and leverage (ND/EBITDA, ND/E, …)

Country

- Developed vs emerging markets

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Using Multiples: Steps

1. Identify peer companies
2. **Compute multiples for peers**
3. Compute average multiples
4. Apply the average multiple to the target
5. Compute target's EV and EqV

Numerator

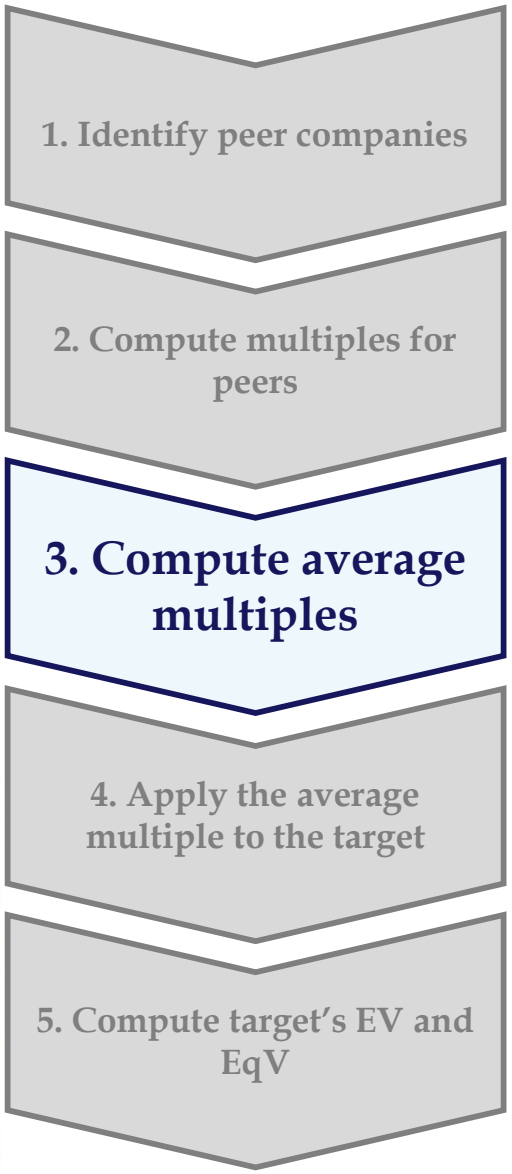
- **Market Cap**: spot price vs 6/12/… month-average
- **Fully Diluted EqV**: Mkt Cap (share price × shares outstanding) + value of in-the-money options, convertible securities, … (if stock price > strike price)
- Alternatively, estimate fully diluted no. of shares. Use the **treasury stock method** to calculate shares arising from option exercises and **ignore treasure shares**

Denominator

- **In general:**
- Exclude **one-off or non-recurring items**
- Apply **same accounting rules** to every company

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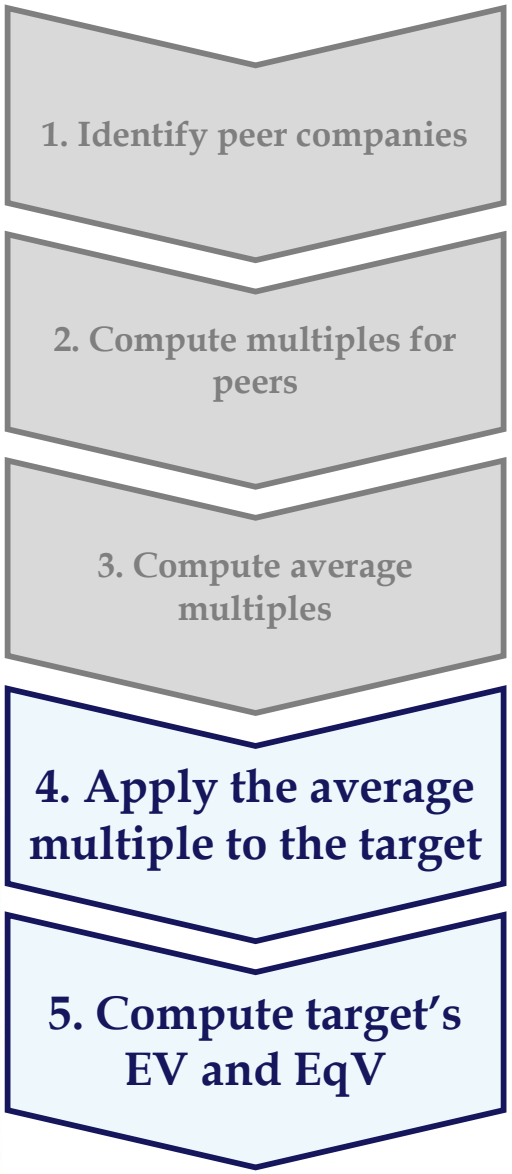
Using Multiples: Steps



Average Multiples

- **Exclude negative multiples and outliers**
- **Possible measures include** mean, median, min, max and a trimmed mean (w/o min and max)
- **Value maps:** Regress peer multiples on a reference variable. Estimate the target's multiple using the regression results
- **Size discount**: If peers are much larger than the target, a size discount may apply

Using Multiples: Steps



Apply Multiples

- **Same adjustments** applied to peers must be applied to the target
- Use a **consistent time horizon**. For example, apply an F+2 multiple to the target's F+2 EBITDA
- Without a business plan for the target, FWD multiples cannot be applied
- For unlisted targets, LTM figures are unavailable if they don't publish quarterly results

Using Multiples: Deal Multiples

- **Deal multiples are retrieved from prices paid in M&A deals:**
  - ✓ Deals from the last 3–5 years are usually considered
  - ✓ They are typically based on actual or LTM financial data
- **The challenge lies in obtaining reliable data about private companies that have been subject of M&A deals**
- **Prices paid for majority/controlling stakes include a control premiums (vice-versa for minority stakes)**
- **Prices may also include the value of synergies, paid to the seller as part of the premium**
- **To derive equity values, prices must be divided by stakes acquired**
- **Prices may be paid in cash and/or shares: both forms of payment must be included**

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Common Multiples: P/E (Definition)

- **P/E is one of the most used (and misused) multiples:**

$$\frac{P}{E} = Price\ /EPS = Market\ Cap/Net\ Income$$

- **There are variants based on how price and EPS are defined**
  1. Price:
     - ✓ Spot price (current one)
     - ✓ Average price to smooth volatility and one-off events
  2. EPS:
     - ✓ Time: EPS in last LFY (current), in most recent four quarters (trailing), expected in next FY (forward) or in some future year
     - ✓ Share outstanding: Primary EPS vs fully or partially diluted EPS. You should ignore (i.e. subtract from the share count) treasury shares and refer to the fully diluted share count
     - ✓ Composition: Before or after extraordinary items

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Common Multiples: P/E (Definition)

- **When P/E are compared it is difficult to ensure EPS are uniformly estimated for the following reasons:**

1. Acquisitions

   Firms often grow by acquiring other firms and they don't account for acquisitions the same way

   While all companies must use *purchase accounting* and record *goodwill*, there is discretion in asset recognition and impairment that could make a material difference in reported EPS

2. Management options

   Using diluted EPS to estimate P/E might bring shares covered by management options into the multiple (be aware about deep in-the-money or only slightly in-the-money options)

3. Accounting choices

   Firms often have discretion in whether they expense or capitalize items, at least for reporting purposes.

   The expensing of CAPEX gives firms a way of shifting EPS from period to period and penalizes those firms reinvesting more

Common Multiples: P/E (Determinants)

- **Current P/E can be derived from a stable growth DDM:**

$$P_0 = \frac{DPS}{K_e - g_n}$$

- **Substituting DPS = EPS_0 · (Payout):**

$$P = \frac{EPS_0 \cdot Payout}{K_e - g_n}$$

- **Dividing both sides by EPS_0:**

$$\frac{P}{EPS_0} = \frac{Payout}{K_e - g_n}$$

- **Forward P/E can be derived from the same equation as above:**

$$\frac{P}{EPS_1} = \frac{Payout}{K_e - g_n}$$

- **We can state the payout as a function of expected g and ROE:**

$$Payout = 1 - g_n/ROE_n$$

- **Substituting into the above equation, we get:**

$$\frac{P_0}{EPS_1} = \frac{1 - g_n/ROE_n}{k_e - g_n}$$

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Common Multiples: P/E (Determinants)

- **Factors that influence the P/E multiple:**

Factor	Relationship to P/E Ratio
Operating risk	Inverse
Leverage ratio	Inverse
Payout (growth rate being equal)	Direct
ROE (growth rate being equal)	Direct
Growth rate (payout being equal)	Direct
Growth duration	Direct

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Common Multiples: PEG

- **PEG is defined as P/E divided by expected g in EPS:**

$$PEG = PE/Expected\ g$$

- **It is pivotal to be consistent:**
  - ✓ Growth rate. G should be the growth rate in EPS rather than EBIT or other metrics
  - ✓ P/E. P/E should be the **current one** (not forward one) otherwise we would account for growth twice (numerator and denominator)
  - ✓ Growth. PEG should be estimated using the same g estimates for all firms.
- **PEG can be computed only for firms with an expected g rate in EPS:**
  - ✓ Since this growth rate is usually provided by analysts, any firm not tracked by analysts will not have a PEG
  - ✓ Since smaller, younger firms tend not to be followed by analysts, this can create selection bias

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Common Multiples: P/BV (Definition)

- **P/BV is computed dividing the market price by BV of Equity:**

$$\frac{P}{BV} = \frac{Price}{BV\ of\ Equity}$$

- **There is potential for inconsistency:**
  - ✓ Shares. If there are multiple classes of shares, the price per share can be different for different classes and it is not clear how the BV of equity should be apportioned among shares
  - ✓ Preferred stocks. You should **not** include the portion of equity attributable to preferred stocks in computing BV of equity, since the market value of equity refers only to **common equity**
  - ✓ BV of equity. BV of equity gets updated infrequently. Consistency demands you use the same measure of BV equity for all firms
  - ✓ Options outstanding. Need to compute the estimated market value of management options and conversion options and add them to the market value of equity before computing P/BV

Common Multiples: P/BV (Determinants)

- **P/BV can be derived from a stable growth DDM:**

$$P = \frac{DPS}{K_e - g_n}$$

- **Substituting DPS = EPS · (Payout):**

$$P = \frac{EPS \cdot Payout}{K_e - g_n}$$

- **Defining the ROE = EPS / BV Equity$_0$:**

$$P = \frac{BV \cdot ROE \cdot Payout}{K_e - g_n}$$

- **Rewriting in terms of P/BV:**

$$\frac{P}{BV} = \frac{ROE \cdot Payout}{K_e - g_n}$$

- **Last equation can be simplified further relating g to ROE:**

$$g = (1 - Payout) \cdot ROE$$

- **Substituting back into:**

$$\frac{P}{BV} = \frac{ROE - g_n}{K_e - g_n}$$

Common Multiples: P/BV (Determinants)

P/BV of a stable firm is determined by the differential between ROE and Ke

- **If ROE > k_e → Price > BV equity**
- **If ROE < k_e → Price < BV equity**

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Common Multiples: EV/EBITDA (Definition)

- **EV/EBITDA is an asset side multiple computed as follows:**

$$\frac{EV}{EBITDA} = \frac{Market\ Cap + Market\ value\ Debt\ - Cash}{EBITDA}$$

- **Since interest income isn't counted in EBITDA not netting out cash will result in an overstatement of the multiple**
- **EV/EBITDA is useful for firms in sectors that require large CAPEX and long gestation periods (Telco and constructions)**
- **It is widely used for several reasons:**
  - ✓ Negative values. There are fewer firms with negative EBITDA than negative EPS and thus fewer firms are lost from the analysis
  - ✓ Accounting choices. Differences in D&A can cause differences in operating income or net income but will not affect EBITDA
  - ✓ Leverage. This multiple can be compared more easily than other earnings multiples with different financial leverage: the numerator is firm value (EV) and the denominator is pre-debt (EBITDA)

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Common Multiples: EV/EBITDA (Determinants)

- **The determinants of EV/EBITDA can be derived from an FCFO model:**

$$EV = \frac{FCFO}{WACC - g}$$

- **We can rewrite FCFO in terms of EBITDA:**

$$\begin{aligned} FCFO &= EBIT(1-t) - (CAPEX - D\&A + \Delta WC) \\ &= (EBITDA - D\&A)(1-t) - (CAPEX - D\&A + \Delta WC) \\ &= EBITDA(1-t) - D\&A(1-t) - Reinvestment \end{aligned}$$

- **Substituting back:**

$$EV = \frac{EBITDA(1-t) - D\&A(1-t) - RR}{WACC - g}$$

- **Dividing both sides by EBITDA:**

$$\frac{EV}{EBITDA} = \frac{(1-t) - \frac{D\&A}{EBITDA}(1-t) - \frac{RR}{EBITDA}}{WACC - g}$$

Common Multiples: EV/EBITDA (Determinants)

- **Five determinants of EV/EBITDA (ceteris paribus):**
  - ✓ Tax rate

    Firms with lower tax rates should command higher EV/EBITDA
  - ✓ D&A

    Firms that derive a greater portion of EBITDA from D&A should trade at lower EV/EBITDA
  - ✓ Reinvestment

    The greater the portion of EBITDA that needs to be reinvested to generate growth, the lower the value to EBITDA will be
  - ✓ Cost of capital

    Firms with lower costs of capital should trade at much higher EV/EBITDA
  - ✓ Expected growth

    Firms with higher expected growth should trade at much higher EV/EBITDA

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Common Multiples: EV/Sales (Definition)

- **There are two basic revenue multiples:**
  1. P/S (Inconsistent!)
  2. EV/S (Consistent)

$$\frac{P}{S} = \frac{Market\ Cap}{Revenues}$$

$$\frac{EV}{S} = \frac{Market\ Cap + Market\ value\ Debt - Cash}{Revenues}$$

- **EV/Sales is more robust than P/Sales because it is consistent:**
  - ✓ It divides the total value of operating assets by revenues generated by those assets, whereas
  - ✓ P/S divides an equity value by revenues that are generated for the firm, consequently
  - ✓ P/S will yield lower values for more highly levered firms

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Common Multiples: EV/Sales (Determinants)

- **Consider the value of a stable growth firm:**

$$EV = \frac{EBIT(1-t)(1-RR)}{WACC - g_n}$$

- **Dividing both sides by revenue, you get:**

$$\frac{EV}{Sales} = \frac{[EBIT\,(1-t)/Sales](1-RR)}{WACC - g_n}$$

$$\frac{EV}{Sales} = \frac{NOPAT\ margin\ (1-RR)}{WACC - g_n}$$

- **EV/S is determined by NOPAT margin, reinvestment and WACC.**
- **Firms with:**
  - ✓ Higher operating margins
  - ✓ Lower reinvestment rates
  - ✓ Lower costs of capital

  Will trade at higher EV/S

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